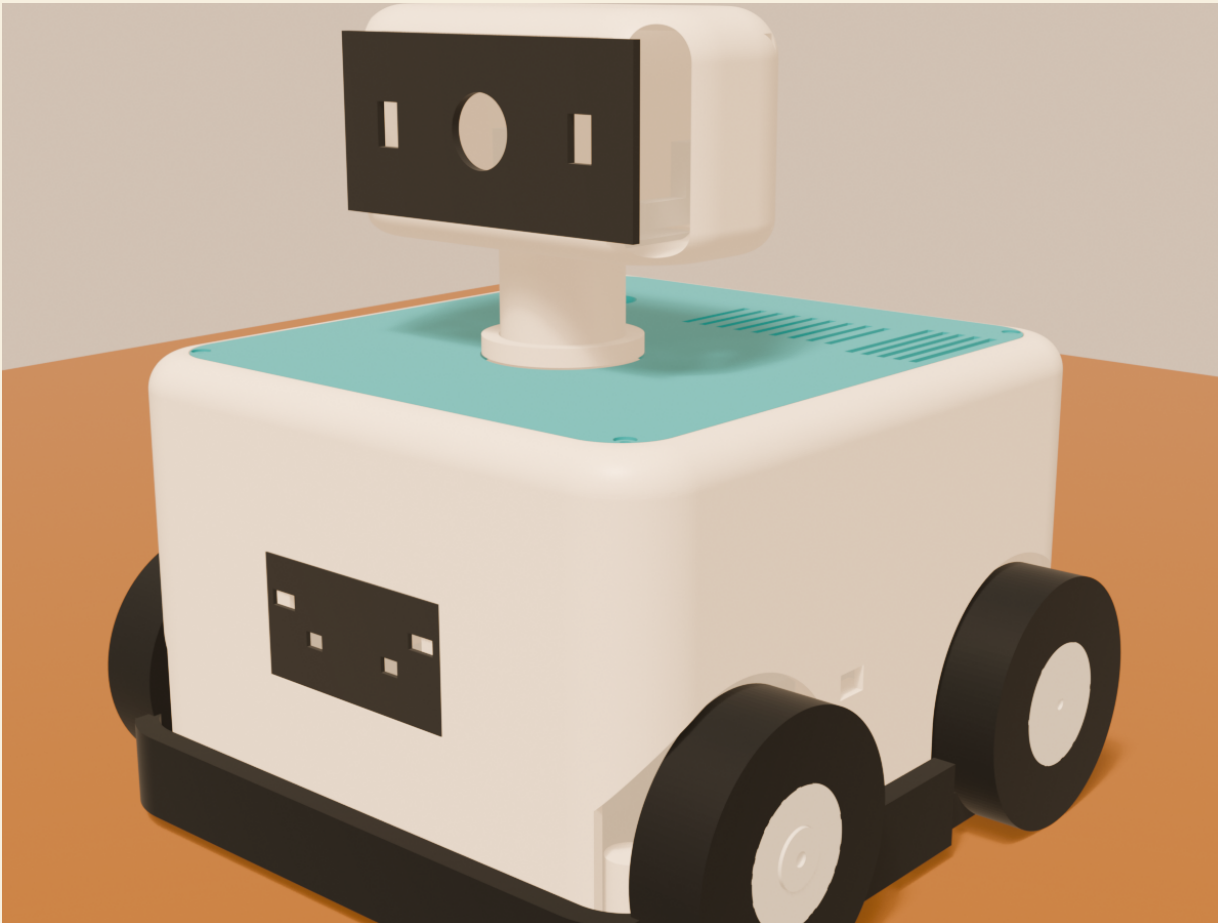




One-piece 238 x 220 x 133 body. No purchased metal structure.

39 printed parts (budget 40).

40 screws + 40 inserts — one SKU each: M3 x 8 socket head cap (ISO 4762); M3 x 5.7 x 4.6 OD heat-set (ruthex RX-M3 class).

One 2.5 mm hex key assembles the robot.

Zero-support print inventory (D028).

Decisions D025-D028; the D027 packing gate is OPEN pending the external review round.

This guide describes the v2 design intent. Nothing here authorizes powered motion; the release gates on the final page hold.

- The physical E-stop cuts motor power independently of the Pi — its dual-NC contacts stay in the low-current relay-enable path.
- Bumper switches and the Pico watchdog stop motion without consulting any software above them.
- Printed structure carries torque and shear through geometry (D-bores, keys, troughs, pockets); screws only clamp (D025/D026).
- Lift the unpowered robot with two hands under the tray. Never lift by the shell, lid, head, bumper, or wiring.
- The E-stop clamps the removable 4 mm lid (inside the IDEC 0.8-6 mm range) against the integrated collar boss; its slap access sweep and the head's pan sweep are validated non-intersecting (finding #11).
- Charge only through the keyed EN2 inlet with charger AC removed; insertion must inhibit motion deterministically.

Every joint: one M3 x 8 socket head cap (ISO 4762) through a 3.0-3.4 mm clamp stack into one M3 x 5.7 x 4.6 OD heat-set (ruthex RX-M3 class). Counterbores keep the stack constant; thicker flanges are counterbored, thinner parts ARE the stack.

Joint family	Screws	Interface points (x, y)
shell_tray	4	(105,96), (105,-96), (-105,96), (-105,-96)
lid_shell	4	(100,90), (100,-90), (-100,90), (-100,-90)
deck_towers	4	(30,-61), (86,-61), (95,61), (107,55)
motor_caps	4	(63,83), (85,83), (63,-83), (85,-83)
battery_clamp	2	(17,42), (17,-42)
controller_tower_base	4	(-101,59), (-101,-59), (-33,59), (-33,-59)
front_pods	4	(-86,85), (-62,85), (-86,-85), (-62,-85)
rear_wheel_clamps	2	(74,118), (74,-118)
front_axle_retainers	2	(-74,126), (-74,-126)
mic_cradle	2	(30,-25), (90,-25)
speaker_clamps	4	(-52,100), (26,100), (-52,-100), (26,-100)
tof_clamps	2	(-32,100), (-32,-100)
pico_clamp	2	(3,43), (3,84)

coupon_insert_m3 [PETG] — insert fit: drive one insert per bore, pick the station that seats flush without melt squeeze-out

coupon_joint_flange [PETG] — with coupon_joint_boss: M3x8 through the counterbored flange; verify 3.2 clamp, full seat, no bottoming

coupon_joint_boss [PETG] — mate of coupon_joint_flange

coupon_dbore_torque [PETG] — mount on the #4867 shaft: hold 2x extrapolated stall (22 kg-cm) without slip or creep, warm and cold (D025 gate)

coupon_axle_stub [PETG] — with coupon_bushing_ring: spin under ~1 kg radial load; measure slop growth over a carpet-equivalent drag distance (D025 gate)

coupon_bushing_ring [PETG] — wear ring for coupon_axle_stub; grease PETG-safe

coupon_snap_pair [PETG] — 10 insert/release cycles; hook must survive and retention must stay positive

coupon_bayonet_pair [PETG] — head-retention interface: engage/disengage feel, no ratcheting under axial pull

coupon_switch_pocket [PETG] — seat one D2HW switch; verify the 0.4 rest gap / 2.0 stroke / 2.4 stop interface before committing the tray print

coupon_tire_fit [TPU+PETG] — print the ring in TPU, the core in PETG; calibrate tire stretch and seat retention at 40% scale

coupon_pcb_clamp [PETG] — drop in the Pico 2, verify peg engagement without board stress, clamp bar seats at 3.2 stack

Design	Qty	Material	Print span (mm)	Orientation
battery_clamp	1	PETG	16.0 x 99.0 x 3.2	flat
battery_pad_frame	1	TPU	96.0 x 56.0 x 1.0	flat
bayonet_collar	1	PETG	52.0 x 52.0 x 6.0	flat
bumper_front	2	TPU	128.0 x 238.0 x 28.0	flat, open-bottom U
bumper_rear	2	TPU	128.0 x 238.0 x 28.0	flat, open-bottom U
controller_tower	1	PETG	99.0 x 126.0 x 41.3	flat
deck	1	PETG	93.8 x 134.0 x 13.0	flat, ribs up
eye_diffuser_bar	1	PETG-lime	30.0 x 18.0 x 4.6	flat
fascia	1	PETG-dark	39.4 x 75.4 x 1.5	flat
front_pod_left	2	PETG	40.0 x 43.0 x 49.0	inboard face down (axle rises vertical)
front_pod_right	2	PETG	40.0 x 43.0 x 49.0	inboard face down (axle rises vertical)
front_wheel	2	PETG	42.0 x 42.0 x 24.0	axis vertical
head_faceplate	1	PETG-dark	56.0 x 120.0 x 3.0	flat, face down
head_pan_plate	1	PETG	42.0 x 74.0 x 3.0	flat
head_shell	1	PETG	72.0 x 140.0 x 71.8	open face down
lid	1	PETG-dark	215.5 x 195.5 x 10.0	top face down

Design	Qty	Material	Print span (mm)	Orientation
mic_cradle	1	PETG	76.0 x 76.0 x 3.2	flat
motor_cap	2	PETG	28.0 x 26.0 x 6.0	flat
neck	1	PETG	40.0 x 40.0 x 44.0	flange down
pico_clamp	1	PETG	10.0 x 49.0 x 3.2	flat
printed_washer	1	PETG	8.0 x 8.0 x 1.6	flat
rear_panel	1	PETG	49.4 x 103.4 x 1.5	flat
rear_wheel	2	PETG	42.0 x 42.0 x 24.0	axis vertical
shell	1	PETG	238.0 x 220.0 x 133.0	upright, open bottom down
speaker_clamp	2	PETG	86.0 x 8.0 x 3.2	flat
status_diffuser_bar	1	PETG-lime	76.0 x 12.0 x 3.2	flat
tilt_bushing	1	PETG	9.2 x 9.2 x 3.6	axis vertical
tire	4	TPU	86.0 x 86.0 x 24.0	flat ring
tof_clamp	2	PETG	20.0 x 8.0 x 3.2	flat
tray	1	PETG	238.0 x 220.0 x 61.0	flat, bottom down
yoke	1	PETG	43.0 x 82.0 x 26.5	TBD (BREP)

1. Print and pass the coupons — All 11 coupons print support-free. Do not print any large part until the insert, joint, D-bore, axle, snap, switch-pocket, and tire coupons pass.

joints: [coupons](#)

2. Heat-set the tray and shell inserts — Drive the M3 inserts flush: tray (controller base, pods, battery posts, Pico bosses, saddle tops, deck towers) and shell (corner lugs, lid ledge, speaker columns, ToF bosses).

joints: [shell_tray](#), [deck_towers](#), [motor_caps](#), [battery_clamp](#), [controller_tower_base](#), [front_pods](#), [speaker_clamps](#), [tof_clamps](#), [pico_clamp](#)

3. Mount the rear motors and wheels — Drop each #4867 into its saddle trough, fit the clamp cap (2 screws each), route the encoder leads into the motor-lead corridor. Press each wheel's D-bore onto the shaft and fit the radial clamp screw.

joints: [motor_caps](#), [rear_wheel_clamps](#)

4. Fit the front pods and wheels — Bolt each pod's base flange to the tray (2 screws), slide the front wheel onto the printed axle, retain with a printed washer + screw into the axle-end insert.

joints: [front_pods](#), [front_axle_retainers](#)

5. Install the controller tower — Bolt the plinth through its four counterbored tabs, seat the MDDS10 on its standoffs (terminals NORTH into the side corridor), stack the Pi on the shelf frame with the AI-HAT reserve above.

joints: [controller_tower_base](#)

6. Seat the battery and Pico — TPU pad frame on the tray pads, pack on top (leads rearward through the corridor and riser), clamp bar onto the rail posts. Pico onto its bosses, clamp bar over it (USB east, SWD up).

joints: [battery_clamp](#), [pico_clamp](#)

7. Hang the regulators and fit the deck — Regulators hang under the deck (D36 wire corridor WEST), fuse block into its pocket lip on the deck's south half, wire exit over the edge. Deck onto the four towers.

joints: [deck_towers](#)

8. Fit speakers, ToF boards, and the shell — Speakers onto the shell ledges with clamp bars; side ToF boards behind their windows with clamps. Lower the shell over the chassis and drive the four corner-lug screws up through the tray.

joints: [speaker_clamps](#), [tof_clamps](#), [shell_tray](#)

9. Close the lid with the E-stop — The lid IS the IDEC clamp panel: mount the E-stop through the lid bore against the integrated collar, purchased nut clamps the 4 mm lid. Mic cradle onto its lid bosses. Four lid screws into the ledge.

joints: [lid_shell](#), [mic_cradle](#)

10. Build the head — Neck through the lid bore onto the pan interface, bayonet collar retains it. Yoke on the neck flange, tilt bushing + M3 pivot on the passive side, faceplate into its recess, pan plate closes the underside.

joints: [head_joints are Phase-4 detail](#)

Nothing in this guide is a powered-motion release.

- Every purchased part: delivered-revision fit check before its assembly step.
- D-bore coupon holds 2x extrapolated stall (22 kg-cm) warm and cold before powered motion.
- Axle/bushing wear test on target flooring before household release.
- Battery clamp inverted-shake and tilt tests; pack current and runtime tests per the BOM.
- E-stop dual-NC, deterministic motor-cut, and bumper six-zone tests before any powered motion.
- Delivered-part weighing replaces every screen-grade mass in the inventory.
- A fresh external design review round precedes the D027 gate declaration.

